

CPSC 250 – Python for Data Manipulation – Summer 2026 Test 1

Topics: Primitive Data Types, Lists & Tuples, Branching, Loops

Part I – Multiple Choice (10 points)

Circle the best answer. Each question is worth 2 points.

1. What is the result of `17 % 5`?
 - A. 3
 - B. 2
 - C. 5
 - D. 17
2. Which of the following creates a tuple?
 - A. `[4, 5, 6]`
 - B. `{4, 5, 6}`
 - C. `(4, 5, 6)`
 - D. `"4, 5, 6"`
3. What is the result of `float("7") + 2`?
 - A. 9
 - B. 9.0
 - C. "72"
 - D. Error
4. Which condition checks whether `age` is less than 18 or greater than 65?
 - A. `age < 18 and age > 65`
 - B. `age < 18 or age > 65`
 - C. `18 <= age <= 65`
 - D. `not age`

5. What is printed by the following code?

```
data = [10, 20, 30]
print(data[-1])
```

- A. 10
- B. 20
- C. 30
- D. Error

Part II – Find the Error (15 points)

Each of the following code snippets contains one or more errors. Identify and correct them.

6.

```
scores = [85, 90, 95]
print(scores(1))
```

7.

```
if x > 10
    print("Large")
```

8.

```
total = 0
for i in range(5):
    total += i
print(total)
```

Part III – Code Writing (20 points)

Write Python code for each of the following tasks.

9. Write Python code that asks the user to enter 8 integers. Count how many of the numbers are even, then print the count.

10. Write Python code that uses a loop to print all multiples of 5 from 5 through 50 inclusive, each on its own line.

Part IV – Code Comprehension and Commenting (15 points)

The following program is complete but has no comments. Write comments to explain what each line does.

```
# _____  
  
numbers = [4, 8, 1, 6, 9]  
  
# _____  
  
largest = numbers[0]  
  
# _____  
  
for n in numbers:  
  
# _____  
    if n > largest:  
        largest = n  
  
# _____  
  
print("Largest value:", largest)
```

Part V – Short Answer (10 points)

11. Explain the difference between `=` and `==` in Python. Give an example of each.

12. What is the difference between a `for` loop and a `while` loop? Describe a situation where each would be appropriate.

Total: _____ / 70